OpenScope

Open-source GLS10 Interoperability

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Abstract

OpenScope is an independent, open-source desktop sonar viewer for receiving, displaying, and recording data from Garmin GLS10 hardware. The observations reported here were made with a GLS10 connected to a Garmin Panoptix LVS32 transducer. The project is free to use, with source code and desktop builds available from github.com/mahobley/OpenScope. This report records what we observed and learned while implementing the display-side network interactions: discovery, session establishment, stream activation, controls, frame reconstruction, and selected telemetry. It distinguishes experimentally necessary behaviour, qualified simplifications, and unresolved fields. Results apply to the tested hardware, firmware, and lifecycle conditions; they are not a public Garmin protocol specification or a guarantee of compatibility with every GLS10 configuration.

1 Project purpose and availability

OpenScope provides a computer-based interface for sonar visualisation, recording, and experimentation with compatible GLS10 hardware. It implements the *display-side peer* role: announcing presence, discovering services, establishing control sessions, requesting sonar delivery, and processing the received stream. These terms describe network interactions and measured outcomes. Where the report uses "head unit" in endpoint descriptions or packet names, it means that display-side protocol role.

Imaging sonar has previously been used for aquatic research and monitoring, including ecological applications [2].

Source code and desktop builds: github.com/mahobley/OpenScope.
Desktop release downloads: OpenScope GitHub Releases.

OpenScope is open source and free to use. The repository provides the source for inspection, reproducible experiments, and further development, alongside desktop builds. Installation and build instructions are in the repository.

OpenScope is an independent project and is not affiliated with or endorsed by Garmin. Product names identify the hardware and observed protocol context. The implementation was developed by observing network traffic and testing protocol interactions. The code was written independently. Garmin firmware was not extracted, inspected, or decompiled, and Garmin source code was not accessed. This report describes interoperability findings without assigning unsupported meanings to captured bytes.

2 What we observed and learned

- **Sustained delivery depends on session state.** Repeated UDP 50030 presence, retained TCP sessions, TCP 50224 active-state updates, and a data-sink activation path were necessary under the tested conditions. A stream starting once was insufficient evidence of a stable session.

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- **Captured traffic is broader than the necessary interaction.** Several auxiliary announcements, captured delays, identity fields, and a heartbeat could be omitted or changed in qualified profiles. Experiments on individual changes do not automatically qualify their combined use.
- **Control values can depend on the current session.** Range and sonar-state messages use a base returned by the GLS10. A completed socket write alone does not demonstrate that a control took effect; received traffic provides an operational check. Disable behaviour remained intermittent.
- **Images and telemetry travel separately.** Image chunks reassemble into metadata plus a greyscale JPEG. An observed angle table and an empirical range calibration support Cartesian rendering; separate packets supply identified temperature and attitude values.
- **Unknown fields remain unknown.** Framing, offsets, observed values, and successful experiments are documented independently of inferred meanings. Uninterpreted bytes are not assumed to be padding, credentials, or encryption.

3 How to read the protocol record

The detailed sections retain a conservative captured baseline so that experiments can be compared and reproduced. Literal packet names and bytes are evidence of that baseline, not a statement that OpenScope must advertise a captured device identity. Section 12 describes tested substitutions and the minimised implementation. The source code is the reference for the behaviour of any particular desktop build.

Throughout this report, "required" denotes behaviour identified as necessary under the tested conditions, whereas "optional" denotes behaviour whose omission passed the reported tests. "Observed" and "captured" describe recorded traffic or values without establishing necessity. Where necessity has not been tested, it remains undetermined; inclusion in the conservative baseline alone does not establish a requirement. No decoded interpretation is assigned to opaque captured bytes.

4 Observed session workflow and tested requirements

The documented OpenScope implementation operated as a TCP client over a direct wired Ethernet link to a GLS10 connected to a Garmin Panoptix LVS32 transducer. The GLS10/LVS32 hardware arrangement is described in Garmin's installation documentation [1]. The GLS10 transmitted sonar data by multicast. The computer used a single dedicated Ethernet interface, either a built-in port or an external Ethernet connection, with head-unit address `172.16.6.1/16`. In the captured baseline, the address was `172.16.6.0/16`. The interface was not assigned `172.16.3.0`, and its physical MAC address replaced the captured head-unit MAC address. Address and neighbour configuration, as well as packet capture, generally require administrator privileges. Figure 1 summarises the network endpoints and the direction of multicast sonar delivery.

The following list describes the software's session workflow, including local discovery and reception choices. It is not a list of independently proven conditions for the GLS10 to transmit sonar data.

1. Concurrent operations comprised the UDP 50030 presence loop (Section 7), UDP 50051 discovery and replies (Section 8), mDNS/DNS-SD browsing, UDP 50223 multicast reception, and TCP workers.
2. GLS presence was detected from a UDP 50051 message, a valid DNS-SD sink advertisement, or a 50223 packet. Peer traffic supported MAC-address learning and scoped neighbour-entry installation; the captured fallback address was `04:25:e8:68:ef:58`.

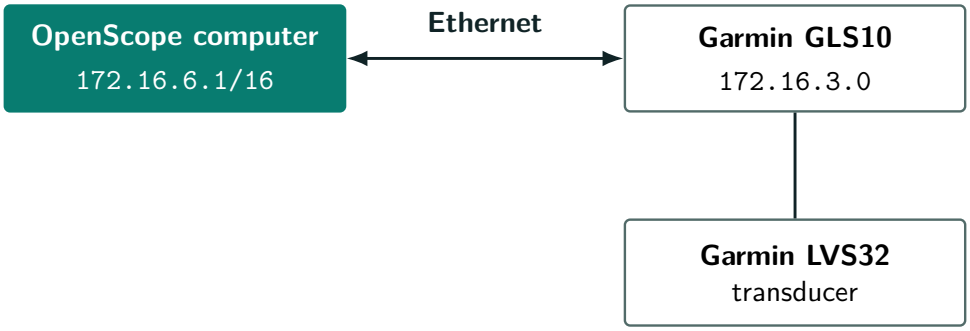


Figure 1: Network configuration used for the documented experiments. OpenScope communicates with the GLS10 over a single Ethernet connection, which carries bidirectional TCP control traffic, UDP discovery and presence messages, and UDP sonar data. The GLS10 transmits sonar data to multicast group `239.254.2.16:50223`, which OpenScope joins to receive the stream. The Garmin LVS32 transducer connects directly to the GLS10.

3. The TCP 50900 socket remained open after initialisation. Subsequent TCP 50224 initialisation provided the current session base at response bytes 12–15, followed by active-state traffic at one-second intervals (Sections 9–11).

4. DNS-SD resolution of `_garmin-ds-sink._udp.local.` supplied the data-sink port, which received an active type-05 request (Section 10).

5. TCP 50227 initialisation preceded the documented enable/disable controls. Multicast group `239.254.2.16:50223` carried the sonar stream.

Under the tested conditions, the delivery-side core comprised repeated UDP 50030 presence, a retained TCP 50900 connection, repeated TCP 50224 active-state updates, and an active data-sink path. TCP 50227 supported control operations. The client used UDP 50051 and mDNS/DNS-SD to discover the GLS10 and resolve its advertised sink endpoint; these implementation choices are distinct from the state needed to sustain sonar delivery. Detection through one channel does not establish that every other detection channel is necessary.

UDP 50223 reception is necessary for OpenScope to reconstruct and display the received sonar data. The tests do not establish that opening the computer's receive socket is itself a condition for the GLS10 to start transmitting. Sections 6 through 12 and Appendix A document these implementation details; Section 12 describes qualified simplifications. Section 13 records frame reassembly, identified telemetry, and polar-to-Cartesian rendering.

5 Session-establishment sequence

```
concurrent UDP/mDNS/receiver workers
          |
 GLS detected + client discovery ready
          |-- (+550 ms captured) --> TCP 50900 init; retain socket
                                              |
                                        (+620 ms captured)
                                              |
                                   TCP 50224 init -> session base
                                              |
                              active state now and every second
                                              |
 DNS-SD sink port -----------------> sink bootstrap/maintenance
                                              |
                                    active type-05 request
```

```
                                  |
(+195 ms captured) --> TCP 50227 init
```

Qualification tests passed with all captured inter-stage delays shown above set to zero. The conservative baseline included 5.5 seconds of sink maintenance before the active request. TCP connection attempts and initialisation reads used a 3-second timeout, with a 2-second retry interval following channel failure. TCP communication was treated as a byte stream: complete records were transmitted, received bytes were accumulated, and `recv` boundaries were not assumed to coincide with record boundaries.

6 Network and socket conventions

The implementation uses IP multicast group membership for sonar reception, following the general conventions of IP multicasting [3].

Hexadecimal strings denote binary octets rather than ASCII text. Explicit integer fields and the apparent TCP length field are little-endian. TCP clients used the selected head address and ephemeral ports. Multicast transmission selected that address as `IP_MULTICAST_IF` with TTL 1; mDNS used TTL 255. Reception enabled reuse, bound wildcard port 50223, joined `239.254.2.16` on the selected head interface, used a one-second timeout, and discarded packets from sources other than `172.16.3.0`.

Table 1: Transport endpoints, captured baseline behaviour, and tested alternatives. Captured baseline details and qualified alternatives are identified separately within each entry.

Channel	Endpoint	Behaviour and qualifications
UDP 50030	head `:50030` to `224.0.0.1:50030`	Repeated 122-byte presence; baseline: ten transmissions at one-second intervals, then every 11.75 s. Qualified alternative: immediate transmission, then every 30 s.
UDP 50051	head and GLS `:50051`	Discovery probes and identity replies until TCP 50224 is ready.
mDNS/DNS-SD	`224.0.0.251:5353`	Browsing `_garmin-ds-sink._udp.local.` and use of the advertised GLS sink port.
TCP 50900	head to GLS `:50900`	Baseline: three records; initialisation-response consumption and continued socket retention and reading. The captured response total of 171 bytes is not a protocol constant.
TCP 50224	head to GLS `:50224`	Baseline: four records; session-base derivation, then active-state transmission every second. Captured response total: 1608 bytes.
Sink UDP	head to discovered GLS port	Initial bootstrap or immediate maintenance, with an active type-05 request.
TCP 50227	head to GLS `:50227`	Baseline: three records before control commands; captured response total: 313 bytes.
UDP 50223	GLS to `239.254.2.16:50223`	Continuous sonar input.

7 Required UDP presence

The `HEAD_50030` record was transmitted from the head address and source port 50030 to `224.0.0.1:50030`. Discovery failed in two tests that used only a single startup datagram. The conservative baseline comprised ten transmissions separated by one second, followed by transmission every 11.75 seconds. A reduced profile that passed qualification testing transmitted immediately and then every 30 seconds. Section A gives the exact 122-byte payload.

8 Discovery, identity, and DNS-SD

Service discovery uses multicast DNS (mDNS) and DNS-based Service Discovery (DNS-SD) [4, 5].

Before TCP 50224 reached the ready state, the following sequence was transmitted every two seconds:

```
9103000000000000
9803000000000000
9603000000000000
9103000000000000
```

For a GLS record beginning `9103`, the implementation returned the 124-, 179-, and 12-byte identity records in Section A; for `9603`, it returned the 12-byte record. The first `9703` triggered one transmission of `9a03000004000000009b760d`. The captured baseline transmitted the 124/179/12/124/179 sequence approximately 300 ms after the first sink cycle. A single 124/179/12 sequence with zero delay also passed qualification testing.

The conservative baseline registered `_garmin-mrn-snr._tcp.local.` with the captured instance/server `garmin-j6-mudskipper-3527325610.local.`, address `172.16.6.0`, and port 50221. The implementation browsed the sink service identified above and accepted an advertisement containing the GLS IP address. When zeroconf was unavailable, the fallback procedure queried both `_garmin-ds-sink._udp.local.` and `_garmin-data-sink._udp.local.`; this fallback cannot provide an equivalent head identity advertisement.

9 TCP initialisation and liveness

The apparent record framing consists of a 4-byte message identifier followed by a 4-byte payload length. The conservative baseline preserves the headers, `efbe`, and declared lengths. The opaque `3f08` and `a90a` payloads remain undecoded.

Table 2: Conservative baseline TCP initialisation records by destination port.

Port	Records in transmission order

50900	`3f08efbe02000000d4c6`
	`a90aefbe0800000090910075c5292f85`
	`c508efbe00000000`

50224	`3f08efbe020000005bdf`
	`a90aefbe08000000ecf7601ca9c046e8`
	`0409efbe0400000000000400`
	`0409efbe1800000014000100305941 37d0107f097a6a4137265943373a5ca446`

50227	`3f08efbe020000002dc4`
	`a90aefbe08000000e47c30483a89abb6`
	`d307efbe00000000`

The fourth 50224 record represents the inactive state and is transmitted before its response is received. The tested implementation required at least 16 response bytes and extracted the

session base as `B = LE32(response[12:16])`. It then transmitted the following active-state record immediately and at one-second intervals. In two tests in which repeated transmission was omitted, image traffic stopped after 22,336 and 22,407 UDP 50223 packets with payloads larger than 1,000 bytes. The first-to-last nonzero reporting intervals spanned approximately 33 seconds in each trial. These are size-filtered datagram counts, not reconstructed image-frame counts; the filter does not independently classify every packet as image data. No further size-filtered image traffic was observed after re-enable in those trials. The active-state record was:

```
0409efbe18000000140001003059413 7d0107f097a6a41372a5943373a5ca446
```

The following 50227 heartbeat was optional under the tested conditions over the tested lifecycle:
`d107efbe0e0000000401040daaeffa910d12d4222101`

10 Data-sink activation

The port obtained through DNS-SD served as the sink destination. The conservative baseline transmitted the following six-record bootstrap once, then sent all six records as a burst every 100 ms:

```
c505000030000000010100
c50500009000000020701058ce794a803
c505000030000000040100
c505000030000000050100
c505000030000000060100
c50500003000000b0100
```

In cyclic traffic, record two was `c50500009000000020701105 + LE16(counter) + 94a803`; the counter began at `0xe78d`, remained unchanged for ten cycles, and incremented modulo 2^{16}. After 5.5 seconds, the inactive type-05 record in each subsequent burst was replaced by the active request below. The active request was therefore repeated every 100 ms as part of the six-record cycle:
`c50500000e000000050c011709ff9f80d0fa96aa8e01`

The six-record burst and immediate periodic maintenance constitute alternatives that passed qualification testing; simultaneous omission of both activation paths was not qualified. The captured source port was 51401, whereas use of an ephemeral source port also passed qualification testing.

10.1 Sink sequence sent by current OpenScope

The current default profile waits for TCP 50224 to be ready, then sends the following bootstrap records consecutively to the GLS10 UDP port discovered through DNS-SD, using an ephemeral source port:

```
c50500003000000b0100
c505000030000000060100
c505000030000000050100
c505000030000000040100
c50500009000000020701058ee794a802
c505000030000000010100
```

About one second later, it sends the following burst and repeats it once per second while the sink worker is running:

```
c50500003000000b0100
c505000030000000060100
c50500000e000000050c011709ff9f80d0fa96aa8e00
c505000030000000040100
c50500009000000020701058de794a802
c505000030000000010100
```

The active type-05 request is the third record in every repeated burst. There is no additional 5.5-second activation delay. The type-02 bootstrap counter is `0xe78e`; the repeated counter is fixed at `0xe78d`. The record order, type-02 suffix, and active type-05 final byte differ from the conservative captured baseline.

This sequence describes the current implementation, not a proven minimum for all devices. Periodic-record omissions were qualified in experimental profiles, but the current reversed-order send path retains all six periodic records. Those omission results should not be read as the packet schedule actually sent by the default runtime.

11 Session-dependent control messages

All control operations used existing sockets in the ready state. Let $S =$ `0x42465312` and $R = (B - S) \bmod 2^{32}$. A range-write record begins with `0409efbe1400000010000200` and contains LE32 words at offsets 12, 16, 20, and 24. The initial words were `14534642`, `47534642`, the chosen range selector, and `16534642`, respectively. The words at offsets 12, 16, and 24 received R. For `min`, the word at offset 20 received the same treatment. For numerical ranges, the selector's bank byte and low 24 bits were relocated independently:

$$\begin{aligned} H &= ((B >> 24) + (W >> 24) - (S >> 24)) \bmod 256, \\ L &= ((B \,\&\, \texttt{0xffffff}) + (W \,\&\, \texttt{0xffffff}) \\ &\quad - (S \,\&\, \texttt{0xffffff})) \bmod 2^{24}, \\ W' &= (H << 24) \mid L. \end{aligned}$$

Table 3: Captured range-selector bytes. Numerical ranges are in feet; `min` is a separate selector.

Range (ft)	Bytes	Range (ft)	Bytes	Range (ft)	Bytes	Range (ft)	Bytes
min	81650981	7	f9df4e82	9	a9e37582	10	81658982
15	e5a0d882	20	81650982	25	1c2a3982	30	e5a05883
35	33037083	40	81658983	45	cfc7a183	50	1c2ab983
55	be6fcc83	60	e5a0d883	65	0cd1e483	70	3303f083
75	5a34fc83	80	81650983	90	cfc62083	100	1c2a3983
110	be6f4c84	120	e5a05884	135	9fea6a84	150	5a347c84
175	3baf9b84	200	1c2ab984				

The sonar-state record on 50224 used the prefix `0409efbe1400000010000200`, followed by `LE32(B)`, `LE32(B+12)`, `LE32(B+d)`, `LE32(B+1)`, where $d = 0$ enabled and $d = 1$ disabled. The associated 50227 records were:

```
enable:  d207efbe0a00000010707010201 00a90100
disable: d207efbe0a00000010707010201 00a90101
```

The conservative sequence transmitted the state record before the command; the enable operation waited for a 50224 response or 50 ms. Successful completion of a write did not, by itself, demonstrate the intended operational effect. Disable operations had historically been unreliable, and 50223 traffic served as the operational measure.

12 Portability and capture-specific parameters

The portable profiles generated identity values once per process, retained them throughout the session, and updated the enclosing length fields accordingly. The tested mDNS name had the form `device-j6-<10 lowercase>-<10 digits>.local.`.

Other tested substitutions included the service `_device-mrn-snr._tcp.local.`, advertised port 50222, a 1–40-byte model string comprising uppercase letters and digits (updated in both

identity records), same-length software/Ilitek/map text replacements, a learned GLS MAC address, head IP `172.16.6.1`, zero inter-stage delays, zeroed opaque initialisation payloads, and omission of the 50227 heartbeat. Captured values provide the conservative comparison baseline for qualification on new hardware. The following components were optional under the tested conditions: raw Ethernet EtherType 0x2003 presence, ARP after learning the neighbour, UDP 50220, 50050, 51000, 51400, 51950, 50615, 8322, UDP 50010 transition, and periodic 50227 heartbeat. In contrast, UDP 50030 remained necessary under the tested conditions.

12.1 minimised OpenScope implementation

The release runtime uses the consolidated minimised profile by default. Compared with the captured baseline, it removes model, software, touchscreen-firmware, and map text from discovery records and repairs the enclosing lengths. It omits the display-side mDNS identity advertisement while retaining DNS-SD browsing for the GLS10 sink. It uses the computer's Ethernet interface MAC address and learns the GLS10 neighbour address from received traffic.

The consolidated profile retains repeated UDP 50030 presence and TCP 50224 active-state updates, an initial sink bootstrap, and repeated sink bursts at one-second intervals as detailed in Section 10. It omits the initial `3f08` records and retains the `a90a` records. These distinctions matter when comparing the literal baseline in Section A with current source code.

Some capture-derived constants remain. In particular, UDP 50030 contains a repeated six-byte MAC-shaped value; replacing it with either the Ethernet interface MAC or a generated local value prevented discovery in the documented experiment. Its protocol meaning has not been established. Keeping those payload bytes is separate from the Ethernet source address, which is the computer's Ethernet interface address. The retained `a90a` payloads also remain uninterpreted. A compatibility result supports retaining a value under tested conditions without establishing its semantic meaning.

Profile notes and experiment records in the OpenScope repository document removals, retained constants, and qualification outcomes. Historical profiles in this section describe stages of investigation rather than a single mandatory packet sequence.

13 Sonar-frame reconstruction and identified telemetry

Two payload classes have been observed in the UDP 50223 stream: chunked image-frame data and independent telemetry records. Image-frame reconstruction precedes JPEG decoding, whereas temperature and attitude values are extracted from separate non-image packets. The field names used in this section describe their observed roles in the implementation rather than an established vendor-defined schema. Figure 2 shows the OpenScope interface displaying reconstructed sonar alongside an echogram and session controls.

13.1 Reassembly of chunked JPEG frames

An image chunk begins with seven little-endian 32-bit words (28 bytes). The implementation accepted only record type `0x00000904` and, by default, the stream ID `0x00060044`. Chunks are grouped by `(stream_id, frame_id, total_len)` and their payloads are placed into a `total_len`-byte buffer at the declared offsets. A frame is considered complete only when every byte in the buffer has been received. The reassembled buffer contains a pre-JPEG metadata block followed by a JPEG image beginning with `ffd8` and ending with `ffd9`. Bytes following the first end marker are excluded from the decoded image.

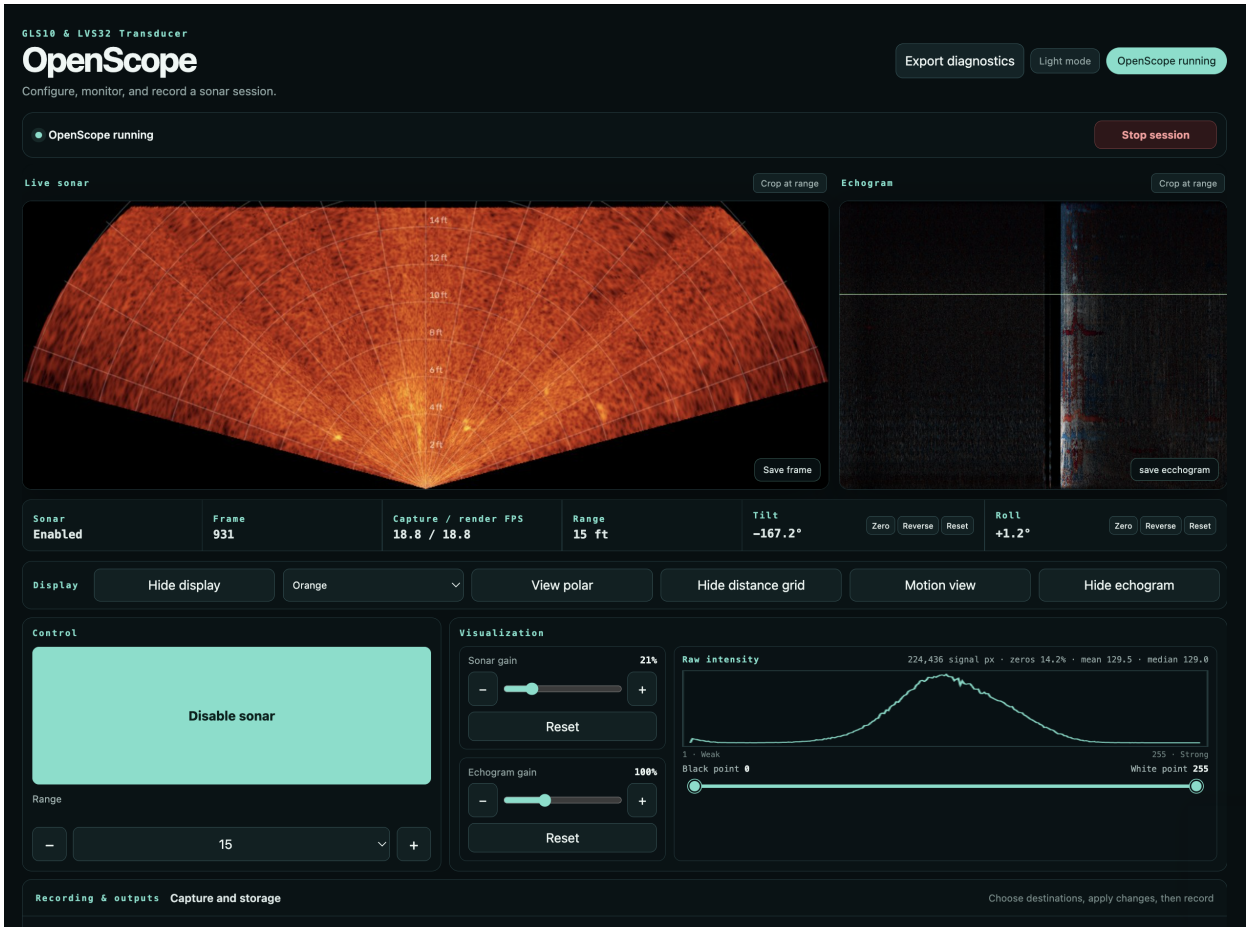


Figure 2: OpenScope during an active GLS10/LVS32 sonar session, showing the reconstructed live sonar view (left) and echogram (right) at a selected range of 15 ft. The orange sonar palette and distance grid are enabled; session telemetry, gain controls, and the raw-intensity histogram appear below. Recording options are located farther down the interface, below the area shown in this screenshot.

Table 4: Observed UDP image-chunk header.

Offset	LE32 field	Recorded interpretation
0	Record type	Frame-chunk discriminator; accepted value `0x00000904`.
4	`len_after8`	Observed field; no independent semantic interpretation has been established.
8	`stream_id`	Image-stream selector; the default filter was `0x00060044`.
12	`frame_id`	Frame sequence identifier used in the reassembly key.
16	`total_len`	Total byte count of the reassembled frame buffer.
20	`offset`	Byte offset of the chunk payload within the frame buffer.
24	`chunk_len`	Number of payload bytes immediately following the header.

The extracted JPEG was decoded as an 8-bit greyscale image. Sampled frames measured 512 by 511 pixels (width by height); these dimensions are empirical observations rather than format constants. In the decoded image, rows correspond to the angular-sample axis and columns to

the range-sample axis. The display pipeline rotates the decoded raster 90 degrees anticlockwise, producing rows indexed by range and columns indexed by angle.

13.2 Per-frame polar metadata

The pre-JPEG block contains an angular lookup table and a range-scale value. Let N_θ denote the decoded JPEG height. A sequence of N_θ consecutive little-endian `float32` values begins at byte offset 68. The observed sequence was monotonic, with values bounded approximately by $\pm\pi$; the implementation interprets these values as scan angles in radians. A further little-endian `float32` value at byte offset 32 is mapped by the current implementation to the reported selected range:

$$R_{\mathrm{ft}} = 2.221279251\, s + 0.020227821, \qquad R_{\mathrm{m}} = 0.3048\, R_{\mathrm{ft}},$$

where s is the raw value at offset 32. These are the exact coefficients used by the implementation for an empirical calibration. Their printed precision does not establish measurement accuracy to nine decimal places; fitting residuals and measurement uncertainty are not reported here. The rendering pipeline uses the angular lookup table, selected range, decoded image dimensions, and valid range rows inferred by the intensity-based heuristic to map image intensities into a Cartesian raster (Section 13.3). The frame identifier groups chunks during reassembly; it does not determine the display geometry. No separate heading or yaw field has been identified in the available record.

For a pre-JPEG block of length P bytes, the following table accounts for every byte in the observed layout. It distinguishes decoded regions from regions retained without a supported semantic interpretation. The 68-byte prefix defines a structural boundary for this description; it does not imply that all fields within the prefix have been decoded.

Table 5: Complete byte coverage of the observed pre-JPEG block.

Byte range	Representation	Recorded interpretation
0–31	Uninterpreted bytes	Retained pre-JPEG header region. No field-level meaning has been established.
32–35	Little-endian `float32`	Raw range-scale value s. The display software maps it empirically to the selected range using the equation above.
36–67	Uninterpreted bytes	Remaining portion of the observed 68-byte header. No field-level meaning has been established.
$68 - 67 + 4N_\theta$	N_θ little-endian `float32` values	Monotonic angular lookup table in radians; one value for each decoded JPEG row (angular sample).
$68 + 4N_\theta - P - 1$	Uninterpreted bytes, possibly empty	Trailing pre-JPEG region before the JPEG start marker. Its presence and contents have not been assigned a semantic interpretation.

The designation “uninterpreted” does not imply that the corresponding bytes are padding or lack protocol significance. It indicates only that the current decoder neither reads nor depends on these regions.

13.3 Polar-to-Cartesian mapping

After the 90-degree rotation, let source row i represent range and source column j represent angle θ_j. For a selected range extent $R_{\max}$ and N_r valid range rows, the implementation uses

$\Delta r = R_{\max}/\max(N_r - 1, 1)$. The coordinate convention is

$$x = r\sin\theta, \qquad y = r\cos\theta,$$

with positive y rendered towards the top of the output and positive x to the right. For each output pixel at coordinates (x, y), the inverse mapping is

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \operatorname{atan2}(x, y).$$

Interpolation of the angular lookup table yields the fractional source-column index. The fractional source-row index is measured inwards from the last valid range row: $i = i_{\text{last}} - r/\Delta r$. Pixels outside the measured angular sector or range interval are assigned zero; the current renderer uses OpenCV linear interpolation by default [6]. Valid range rows are inferred from intensities above 8 in the central angular column, with a fallback across all angles for sparse scenes. This procedure is a display-geometry heuristic rather than a documented sonar measurement rule. Weak or empty scenes may change the inferred valid-row boundary and therefore the displayed geometry.

13.4 Independent temperature and attitude telemetry

The telemetry decoders identified in the current implementation are summarised below. They operate on separate UDP payloads rather than fields embedded in each JPEG frame. The viewer therefore retains the most recently observed value of each quantity for display alongside subsequent frames.

Table 6: Currently identified non-image telemetry fields.

Quantity	Packet recognition	Extracted value
Temperature	Payload begins `d8070000`; at least 14 bytes available.	LE `float32` at offset 10, interpreted as degrees Celsius.
Tilt axis	Payload length 40 and bytes 8–11 equal `1c000800`.	LE `float32` radians at offset 16, converted to degrees. Its physical sign and zero point are empirical.
Roll/cant axis	Same 40-byte packet recognition as tilt.	LE `float32` radians at offset 20, converted to degrees. Its physical sign and zero point are empirical.

The displayed tilt is a relative angle, calculated by subtracting an operator-selected reference (default 90 degrees) and applying an initially negative sign. The displayed roll is calculated by subtracting a reference (default 0 degrees), with an initially positive sign. These reference and sign conventions are display-calibration choices rather than established wire-format semantics. Other non-image packets are logged using byte, unsigned and signed little-endian 32-bit, and `float32` interpretations for subsequent correlation.

14 Validation methodology and limitations

The validation protocol comprised cold starts, verification that all TCP sessions remained open, confirmation of an active sink request, and observation of sustained 50223 image traffic for at least five minutes. Test records documented GLS firmware and hardware, exact packet differences, time to first image, packet rate, and cold/warm state. Single-factor tests isolated individual changes; combined-profile qualifications then checked selected changes together. Disable behaviour was recorded separately where it was not a qualification gate.

Initialisation-response totals and framing semantics remain empirical observations rather than established protocol guarantees. Length-delimited record parsing, together with validation of the

50224 session-base record, offers a more robust basis for interpreting the observed traffic than reliance on captured byte totals. The reported requirements and qualified simplifications are limited to the GLS10/LVS32 setup and the firmware and lifecycle conditions tested. Other transducers and hardware combinations have not been established as compatible by these results.

14.1 Remaining questions

Remaining questions include retained initialisation and presence fields, reliable sonar-disable confirmation, unresolved pre-JPEG regions, higher-resolution GLS10 output, and calibration and telemetry across other hardware and firmware. The intensity-based valid-range-row heuristic has not been established as a reliable geometric boundary detector under all conditions, particularly weak or empty scenes. These findings do not establish broad compatibility or a vendor-defined schema. Future reports should identify the build, hardware, firmware, changed fields, and measured stream or control outcomes.

15 Conclusion

OpenScope provides an independent, open-source computer-based sonar viewer for the tested GLS10/LVS32 setup. This report documents an experimentally supported subset of display-side interactions and an image reconstruction pipeline. Broader compatibility, unresolved fields, disable reliability, and empirical display geometry remain areas for further validation.

References

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[6] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal*, vol. 25, no. 11, pp. 120, 122–125, 2000.

A Literal packet appendix

The following records constitute the conservative captured baseline packet set, including historical identity text and initialisation values. They are retained to document the observations; the minimised runtime differs as described in Section 12. Whitespace is included solely for readability; each named record is formed by concatenating its hexadecimal lines.

```
HEAD_50030:
2c0a000072000000fffffffffff
148f21304377148f21304377148f21304377148f21304377
148f21304377148f21304377148f21304377148f21304377
148f21304377148f21304377148f21304377148f21304377
148f21304377148f21304377148f21304377148f21304377
474d4e310053595300323000

HEAD_DISCOVERY_124:
92030000740000000200000001005411
4543484f4d415020556c74726132203130367376000001001900000054000100
43484f4d415020556c74726132203130367376000300010016000100080003000
0e0009100000537973746 56d20436f6465000000505bb70b0204030000000a00
000000000000000505bb70b

HEAD_DISCOVERY_179:
9b030000ab000000505bb70b541109109c0000004543484f4d415020556c747261
32203130367376205 36f667477617265205665727 3696f6e2034312e30350a696c
6974656b5f74733a207665722d696c6974656b206669726d776172652076657273
696f6e20697320372e302e302e332e302e302e302e300a564552424d4150204255
533030314c202d20552e532e202620436f617374616c2043616e616461200a5645
52534d4150204e6f6e6500010107

HEAD_DISCOVERY_12 = 9703000004000000aab73ed2
HEAD_DISCOVERY_FINALIZE = 9a03000004000000009b760d
HEAD_STREAM_TRANSITION = 7408000000000000
TCP_50900_1 = 3f08efbe02000000d4c6
TCP_50900_2 = a90aefbe0800000090910075c5292f85
TCP_50900_3 = c508efbe00000000
TCP_50224_1 = 3f08efbe020000005bdf
TCP_50224_2 = a90aefbe08000000ecf7601ca9c046e8
TCP_50224_3 = 0409efbe0400000000000400
TCP_50224_INACTIVE = 0409efbe18000000140001003059413 7d0107f097a6a4137265943373a5ca446
TCP_50224_ACTIVE = 0409efbe18000000140001003059413 7d0107f097a6a41372a5943373a5ca446
TCP_50227_1 = 3f08efbe020000002dc4
TCP_50227_2 = a90aefbe08000000e47c30483a89abb6
TCP_50227_3 = d307efbe00000000
TCP_50227_HEARTBEAT = d107efbe0e0000000401040daaeffa910d12d4222101
SONAR_ENABLE = d207efbe0a00000010707010201 00a90100
SONAR_DISABLE = d207efbe0a00000010707010201 00a90101
SINK_1 = c505000003000000010100
SINK_2_ONE_TIME = c5050000090000000207010 58ce794a803
SINK_3 = c505000003000000040100
SINK_4_INACTIVE = c505000003000000050100
SINK_4_ACTIVE = c50500000e000000050c011709ff9f80d0fa96aa8e01
SINK_5 = c505000003000000060100
SINK_6 = c50500000300000000b0100
```